

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year 2025

Subject: 031001111 Mechanics I

Section 15-17

Date Monday, August 4, 2025

Time 08.00 - 10.00

Examination Setter Name: Asst.Prof. Dr. Dusit Ngamrungraj, Dr. Pinthudit Klinkajorn

Student Name.....ID.....Class.....No.....

Directions:

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 2 sections on 5 pages (including the cover page). The total score is 35 points.
3. **Complete** all questions in the answer booklet.
4. Books, documents and lecture notes are **not allowed**
5. Electronic calculators are **not allowed**
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency
9. **Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.**
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

Cheating in examinations is considered a serious offense and carries the
penalty of expulsion from student status.

This examination is part of the assessment of students' abilities as follows:

		1	2
CLO 1	Describe the principle and concept of measurement, dimension and significant figure.	1-3	1-3
CLO 2	Calculate of measurement discrepancies into account when presenting results. Able to analyze, interpret meaning from straight line graphs and calculate of vector and scalar quantities.	4	4-5
CLO 3	Explain and find the relationship between the position, displacement, velocity and acceleration of an object's with constant acceleration from graphs and equations Including vertical movement under the gravity of the Earth.	5	6-9

Part I Multiple choices.

Choose the best answer for each question (1 point each)

1. Which one is **not** SI base unit?

- a. gram
b. kelvin
c. second
d. ampere

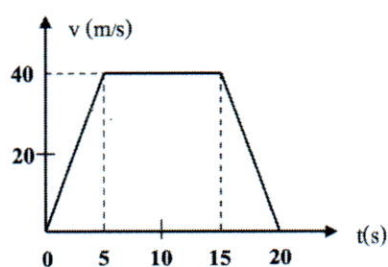
2. The prefix meaning 1000000 is called?

- a. deci
b. mega
c. milli
d. kilo

3. How many significant figures of 0.0080

- a. 1
b. 2
c. 3
d. 4

4. From the graph, find the distance of the movement?



- a. 200 m
b. 400 m
c. 500 m
d. 600 m
5. If a car is travelling at a speed of 90 km/hr, what is this speed in m/s?
- a. 25
b. 90
c. 168
d. 202

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	a	b	c	d
1				
2				
3				
4				
5				

Part II Writing

1. Convert length measurement to the same units. (3 points)

1.1) $4,000 \text{ m} + 3 \text{ km} = \underline{\hspace{2cm}} \text{ m}$

1.2) $700 \text{ kg/m}^3 + 0.8 \text{ g/cm}^3 = \underline{\hspace{2cm}} \text{ mg/mm}^3$

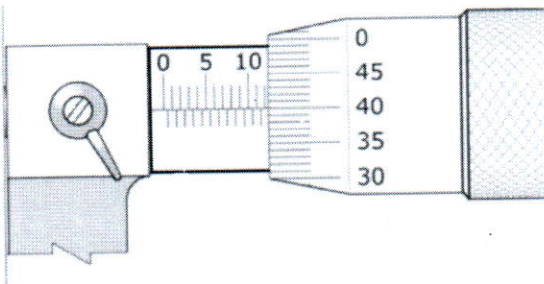
1.3) $72 \text{ km/h} + 10 \text{ m/s} = \underline{\hspace{2cm}} \text{ m/s}$

2. If v and u is velocity, a is acceleration and s is displacement, show that the expression $v^2 = u^2 + 2as$ is dimensionally correct? (3 points)

3. What is the reading value? (2 points)

The value read from the micrometer is

length of arrow is



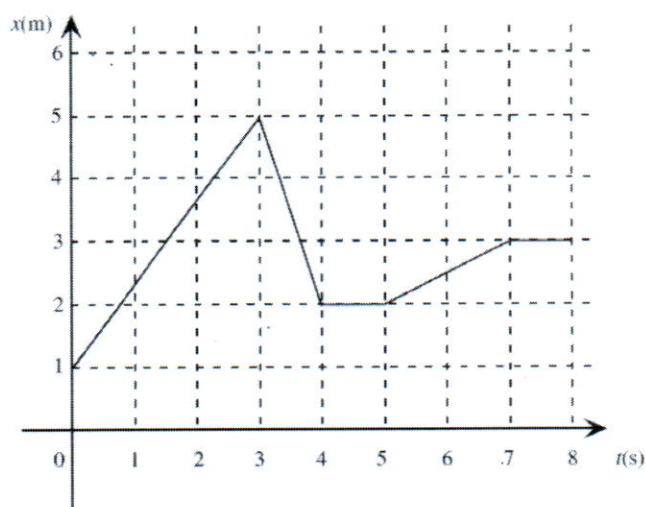
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4. A boy walks along a road at 20 m/s for 3 minutes and then at 10 m/s for 2 minutes. Find his average speed throughout the motion? (3 points)

5. A particle moves in a straight line, shown in a graph of the position of the particle at different times relative to a reference point, where the plus sign represents the right-hand side of the reference point. (10 points)

- a) Find the displacement and distance of this particle from time $t = 0$ s to time $t = 8$ s.
- b) Find the average velocity in the time interval from 0 s to 8 s?
- c) Find the acceleration at time $t = 2$ s?



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6. A race car accelerates uniformly from 18 m/s to 46 m/s in 2 s. Determine the acceleration of the car and the distance traveled? (3 points)

7. A car travels in a straight line on a flat road. It starts to apply the brakes at a constant speed of 36 km/h until it comes to a stop takes 20 seconds. (5 points)

- Find
- a) The total distance traveled until the car stop?
 - b) Acceleration of a car?

8. A man throws a stone straight upward into the air with a speed of 20 m/s from a high cliff. It takes 5 s for the stone to reach the ground. What is its speed and direction 1 s after it is thrown? (3 points)

9. A stone is thrown from the top of a building with an initial velocity 20 m/s straight upward, at an initial height 50 m above the ground. Find

- a) Time needed for a stone to reach its maximum height? (3 points)
- b) The maximum height? (2 points)

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